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Reducing Cognitive Load in Clinical Decision-Making with GenAI-Powered Report Analysis

– a prototype and study proposal

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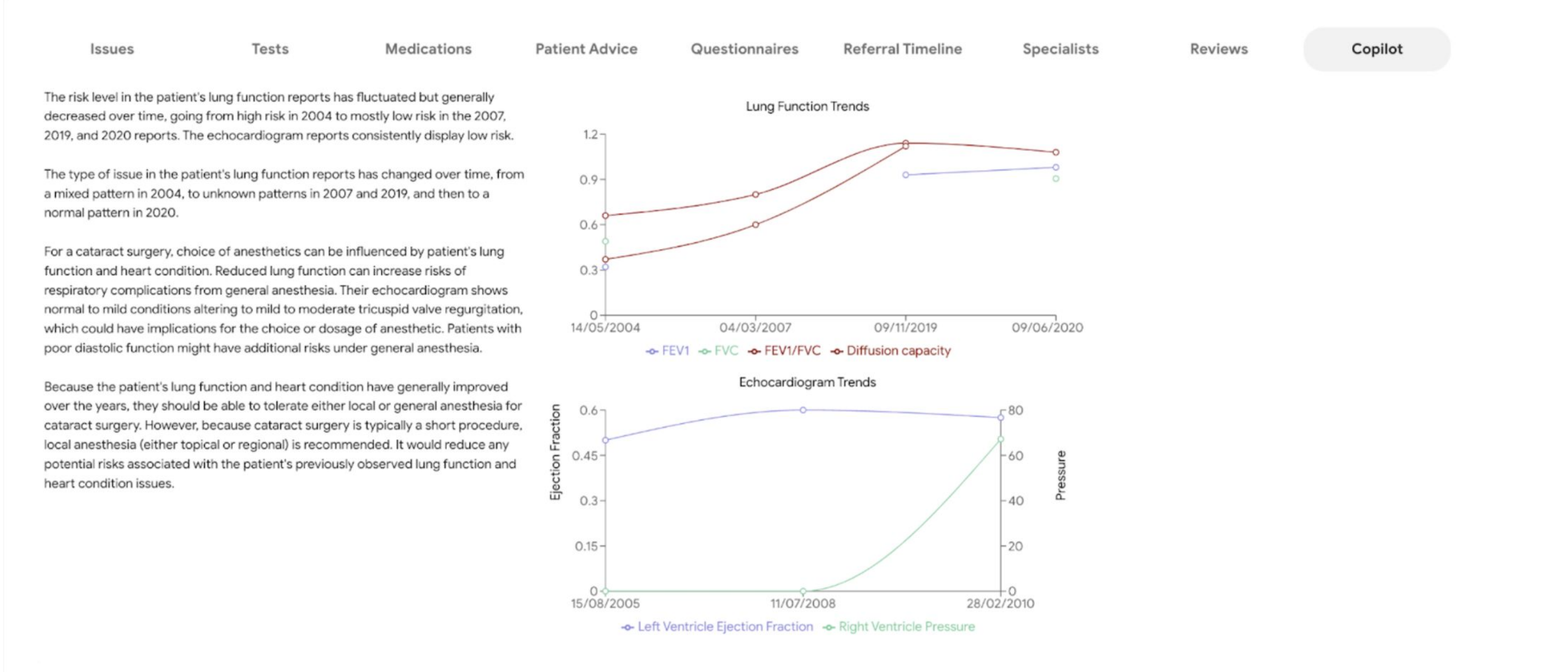
BACKGROUND

Clinicians managing complex patients must manually navigate multiple longitudinal test reports. It is a cognitively demanding process that makes it difficult to identify trends and synthesise findings at the point of care. The burden of integrating these findings can lead to missed insights and suboptimal clinical decisions.

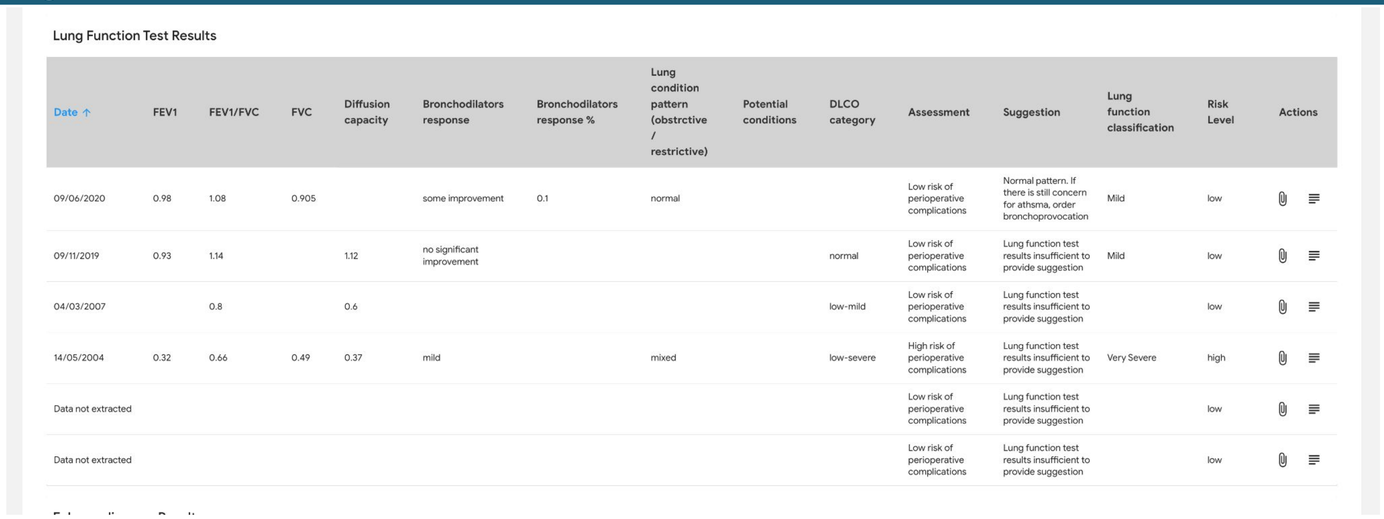
AIMS

To develop a system that automatically extracts, summarises, and visualises trends from longitudinal patient test reports (structured and unstructured data), and surfaces personalised best-practice clinical recommendations.

Report Summary & Trend Visualisation



Structured Longitudinal Results Table

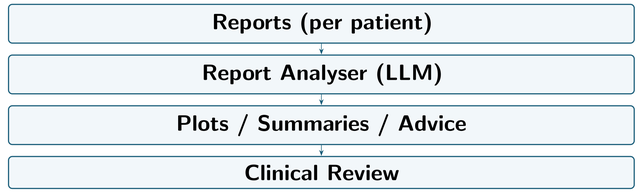


Lung Function Test Results

Date ↑	FEV1	FEV1/FVC	FVC	Diffusion capacity	Bronchodilators response	Bronchodilators response %	Lung condition pattern (obstrctive / restrictive)	Potential conditions	DLCO category	Assessment	Suggestion	Lung function classification	Risk Level	Actions
09/06/2020	0.98	1.08	0.905		some improvement	0.1	normal			Low risk of perioperative complications	Normal pattern. If there is still concern for athsma, order bronchoprovocation	Mild	low	
09/11/2019	0.93	1.14		1.12	no significant improvement				normal	Low risk of perioperative complications	Lung function test results insufficient to provide suggestion	Mild	low	
04/03/2007		0.8		0.6					low-mild	Low risk of perioperative complications	Lung function test results insufficient to provide suggestion		low	
14/05/2004	0.32	0.66	0.49	0.37	mild		mixed		low-severe	High risk of perioperative complications	Lung function test results insufficient to provide suggestion	Very Severe	high	
Data not extracted										Low risk of perioperative complications	Lung function test results insufficient to provide suggestion		low	
Data not extracted										Low risk of perioperative complications	Lung function test results insufficient to provide suggestion		low	

METHODS

The system uses a model-agnostic LLM to iteratively process patients' test reports, extract data points for key clinical metrics, plot trends and create a summary. The prompt specified instructions to parse reports, expected metrics and guardrails to avoid hallucinations. Best-practice recommendations were codified and dynamically presented to the user through a rule-based look-up mechanism, personalised for the patient. The system was tested on respiratory function tests and echocardiograms.



RESULTS

The system analysed reports, extracted key information, presented trends over time in plot and table formats, provided text-based summaries and gave best-practice clinical recommendations. Results were reviewed by an experienced anaesthetist and found to be accurate.

CONCLUSIONS

This proof-of-concept demonstrates that a GenAI-based system can accurately synthesise longitudinal patient reports, visualise results and surface personalised best-practice recommendations, significantly reducing clinician cognitive load. A prospective clinical trial is warranted.

FUTURE WORK

- RAG-based dynamic best-practice guideline lookup
- Clinician chat interface for interactive report discussion
- Prospective clinical trial evaluation

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